

Research Topic: Initial Value Problem

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MATH 4500 Final Project

ABSTRACT

We are targeting initial value problems for ordinary differential equations by implementing numerical methods to find patterns and solve problems. Modern science and engineering are modeled by differential equations. Differential equations have been long studied with large breakthroughs in the 1670's and 1680's from Newton, Leibniz, and the Bernoulli brothers. Newton's work established ordinary differential equations and partial differential equations. Leibniz implemented the use of an integral for solving differential equations. Orthogonal trajectories and isoperimetric problems were studied and established by the Bernoulli brothers. These findings lay the backbone of our knowledge of differential equations today and allow us to use various numerical methods to solve initial value problems.

INTRODUCTION

We utilize the logistic model developed by P.F. Verhulst in 1838 which models population growth over time, $y' = ky(L - y)$. This model accounts for growth/loss due to various factors, rather than the initial logistic model which resulted in exponential growth without limits, $y' = ky$. We have that 'y' represents the number of individuals in the population/group that are infected/having the information and 'L' is the constant size of the group. We notice the behavior of the logistic function is extremely sensitive with respect to the value of the coefficient k and the constant L.

Epidemiology models are used to study the spread of diseases with differential equations. In 1927, Kermack and McKendrick developed the first of these types of models. The Kermack-McKendrick model is an “SIR Model for the number of people infected with a contagious illness in a closed population over time” (Weisstein, Eric W.). This model is meant to observe and understand the large increases and decreases in the number of infected patients. The Kermack-McKendrick model holds assumptions that: the population size is fixed so that there are no changes in the population due to births and deaths from other causes, and there is no immigration or emigration, β and γ are constants, and the population is well-mixed, so every individual meets every other individual with equal probability. With these assumptions, we divide the population into three groups: The susceptible population $S(t)$, the infected population $I(t)$, and the recovered population $R(t)$. We derive the differential equations,

$$\begin{aligned}\frac{d}{dt}S &= -\beta SI, S(0) = S_0, \\ \frac{d}{dt}I &= \beta SI - \gamma I, I(0) = I_0, \\ \frac{d}{dt}R &= \gamma I, R(0) = R_0.\end{aligned}$$

with the assumption that the rate at which infected individuals recover is proportional to $I(t)$.

We study traffic flow, which focuses on the behavior and movement of vehicles, their drivers, and environmental factors. We use a microscopic model of traffic flow which “attempts to analyze the flow of traffic by modelling driver-driver and driver-road interactions within a traffic stream which respectively analyses the interaction between a driver and another driver on road and of a single driver on the different features of a road” (Tom V. Mathew). We derive the car-following model that describes how driver behavior is influenced,

$$\frac{d^2x}{dt^2} + \beta\tau \frac{dx}{dt} + \beta x = \beta y$$

where ' β ' equals the sensitivity coefficient, ' d ' represents the preferred separation between the two vehicles, ' x ' models the position of the following car, and ' y ' denotes the position of the lead car.

NUMERICAL SCHEME

Problem 1. We are given the data that an initiative group of 10 students wants to organize a concert of a famous rock band on campus to celebrate the upcoming graduation. While negotiating with the band's manager, they want to keep the preparation a secret from the other 2,000 students. However, the rumor begins spreading.

Questions:

- A. Determine the coefficient k conditioning the spread of the rumor if all students on campus know about the negotiations in 20 days.
- B. Assuming that k is under the control of the initiative group, generate the value of this coefficient which would guarantee that by the end of the 30-day organization period only 20 students on campus.

Solution:

- A. We directly implement the logistic method into MATLAB, and then solve it for the coefficient k . This direct approach is more accurate because there is no numerical error, and we derive exact values of k .

Code I.

```
>> clc; clear;

% Known values
L = 2000;      % Total population
y0 = 10;       % Initial students who know
% Problem A: y(20) = 1999
tA = 20;       % days
yA = 1999;     % population at day 20
A = (L - y0) / y0;
% Solve for k using:  $k = - (1 / (L * t)) * \ln( ((L / y) - 1) / A )$ 

kA = - (1 / (L * tA)) * log(((L / yA) - 1) / A);

fprintf('Problem A: k = %.8f\n', kA);
```

Problem A: k = 0.00032234

B. We directly implement the logistic method into MATLAB, and then solve it for k, similarly to Question A.

Code II.

```
>> clc;
clear;

% Known values
L = 2000;      % Total population
y0 = 10;       % Initial students who know

A = (L - y0) / y0;
% Problem B: y(30) = 20

tB = 30;       % days
yB = 20;       % population at day 30

kB = - (1 / (L * tB)) * log(((L / yB) - 1) / A);

fprintf('Problem B: k = %.8f\n', kB);
```

Problem B: k = 0.00001164

Problem 2.

Questions:

- A. Explore the model. Set up different initial conditions and observe what happens in the simulations. Observe the effect of each parameter.
- B. Assume there are 7, 138 people. Initially, one person has been infected. Define your initial conditions and solve your system of differential equations defined above. Assume $\beta = 1/7138$, $\gamma = 0.40$. How many people are infected on day 6? Plot your solution curves, $S(t)$, $I(t)$, and $R(t)$ verses time (days) for the first 40 days on one graph.

Solution:

A. Using Euler's Method, this numerical method allows us to change parameters such as β and γ , as well as initial conditions to observe how the model reacts. We observe that increasing the infection rate results in a larger number of individuals being infected, and at a faster rate. While decreasing the infection rate results in a lower total number of individuals being infected and the peak of infection rates happens at a later time. By increasing the recovery rate, fewer individuals are infected at the peak and the duration of the disease is shorter. Decreasing the recovery rate results in individuals having the disease for longer amounts of time and more individuals are infected at the peak. We observe that increasing the initial number of individuals infected causes a larger, earlier peak of infections, while decreasing the number of initial individuals infected causes the disease to spread slower at the beginning.

B. We use Euler's Method and Runge-Kutta 4th order to approximate the number of people that are infected on day 6. Since the SIR Model is a non-linear system of equations, Euler's Method allows us to use small step sizes to achieve high accuracy over a 40-day period. We conclude that 32.62 people were infected on day 6, thus approximately 32 people. Observing Figure 1, we

see that the peak of infected individuals occurs at approximately day 16. We implement RK4 since it has higher order, and thus more accuracy. Then, using RK4, we observe that 36.12 people were infected on day 6, thus approximately 36 people (Code IV). We notice in Figure 2 that the peak of infected individuals occurs at approximately the same day using Euler's Method and RK4.

Code III.

```
% Parameters
beta = 1/7138; % Infection rate,  $\beta = cp$ ,  $cp$  is a contact rate * infection probability
gamma = 0.40; % Recovery rate,  $\gamma$ 
N = 7138; % Total population

% Initial conditions
S0 = N - 1; % Initially, 1 infected individual and the rest are susceptible
I0 = 1; % Initially, 1 infected individual
R0 = 0; % Initially, no one is recovered

% Time span for the simulation
t_max = 40; % Simulate for 40 days
dt = 0.1; % Time step (in days)
t = 0:dt:t_max; % Time vector

% Initialize arrays to store the results
S = zeros(size(t)); % Susceptible population
I = zeros(size(t)); % Infected population
R = zeros(size(t)); % Recovered population

% Set initial values
S(1) = S0;
I(1) = I0;
R(1) = R0;

% Euler's method loop to solve the system
for i = 1:length(t)-1
    % Compute derivatives
    dS = -beta * S(i) * I(i); %  $dS/dt = -\beta * S * I$ 
    dI = beta * S(i) * I(i) - gamma * I(i); %  $dI/dt = \beta * S * I - \gamma * I$ 
    dR = gamma * I(i); %  $dR/dt = \gamma * I$ 

    % Update values using Euler's method
    S(i+1) = S(i) + dS * dt;
    I(i+1) = I(i) + dI * dt;
    R(i+1) = R(i) + dR * dt;
end
```

Code IV.

```

% RK4 Loop
for i = 1:N
    t(i+1) = t(i) + dt;

    % k1
    dS1 = -beta * S(i) * I(i);
    dI1 = beta * S(i) * I(i) - gamma * I(i);
    dR1 = gamma * I(i);

    % k2
    S2 = S(i) + 0.5 * dt * dS1;
    I2 = I(i) + 0.5 * dt * dI1;
    dS2 = -beta * S2 * I2;
    dI2 = beta * S2 * I2 - gamma * I2;
    dR2 = gamma * I2;

    % k3
    S3 = S(i) + 0.5 * dt * dS2;
    I3 = I(i) + 0.5 * dt * dI2;
    dS3 = -beta * S3 * I3;
    dI3 = beta * S3 * I3 - gamma * I3;
    dR3 = gamma * I3;

    % k4
    S4 = S(i) + dt * dS3;
    I4 = I(i) + dt * dI3;
    dS4 = -beta * S4 * I4;
    dI4 = beta * S4 * I4 - gamma * I4;
    dR4 = gamma * I4;

```

Figure 1.

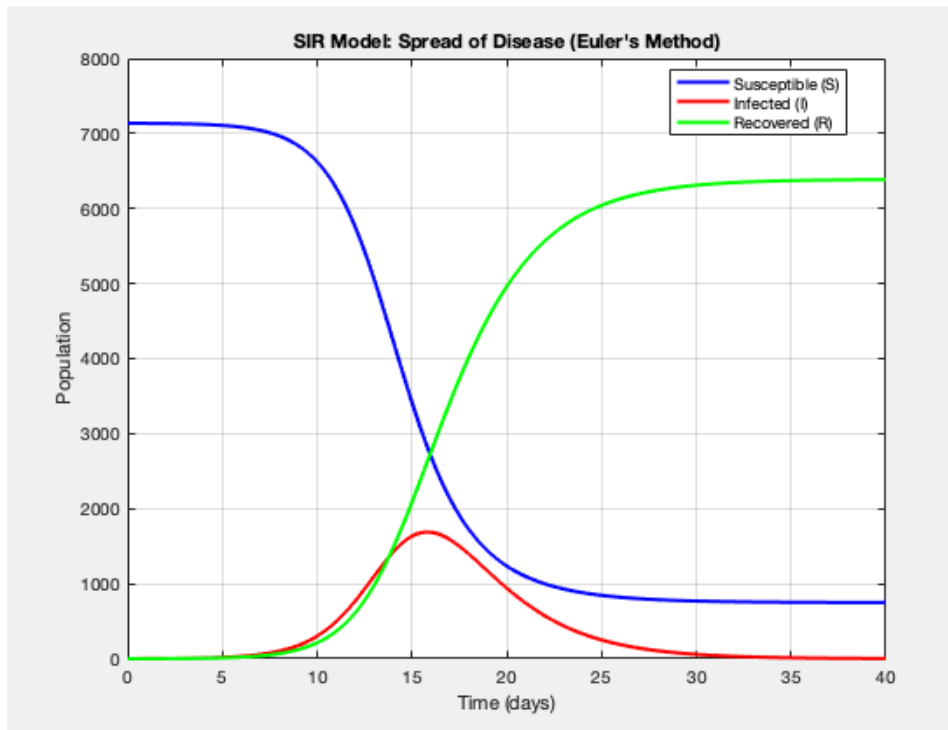
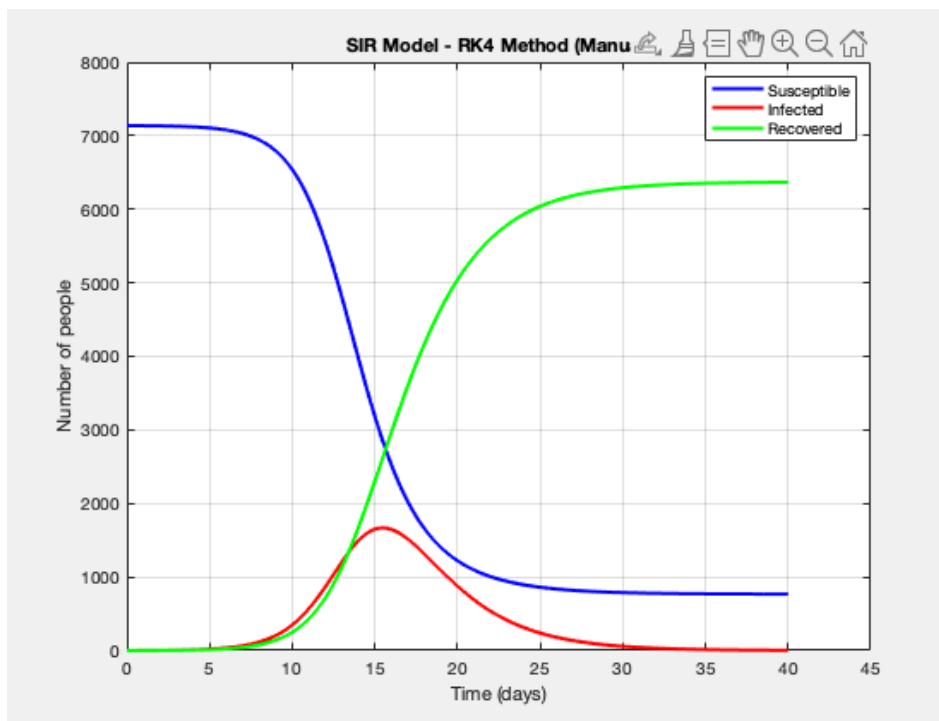


Figure 2.



Problem 3.

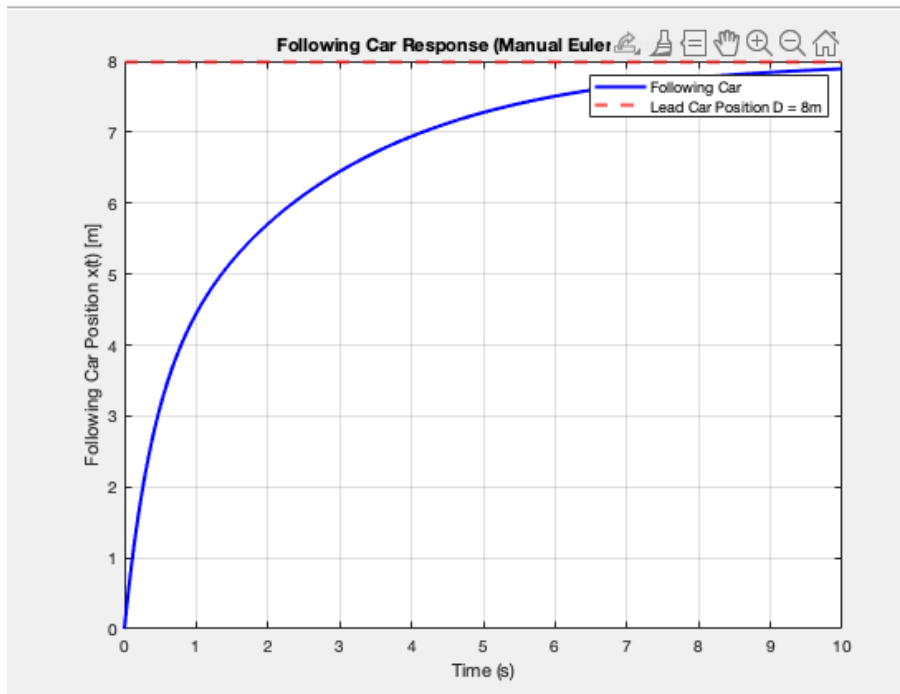
Questions:

- A. Assume $\beta = 1\text{s}^{-2}$, $\tau = 3\text{s}$, $U = 10\text{m/s}$, and $D = 8\text{m}$. In this case solve the ODE and plot the solution with respect to time (s). Do you think it will lead to a collision between the vehicles?
- B. In the same parameter setting, can you predict the required stopping distance to avoid collision?

Solution:

A. We use Euler's Method to solve the ODE and plot the solution with respect to time (s). We chose Euler's Method because it is useful in approximating dynamic systems, such as travelling vehicles. From the plot, I initially guessed that there would be a collision between the vehicles.

Figure 3.



Code V.

```

% Parameters
beta = 1;
tau = 3;
D = 8;
U = 10;

% Time setup
dt = 0.01;           % time step
T = 10;              % total time in seconds
N = T / dt;          % number of steps
t = 0:dt:T;          % time vector

% Initialize vectors
x1 = zeros(1, length(t)); % position x
x2 = zeros(1, length(t)); % velocity dx/dt

% Initial conditions
x1(1) = 0;           % x(0)
x2(1) = U;           % dx/dt(0)

% Euler's method loop
for i = 1:N
    dx1 = x2(i);
    dx2 = -3 * x2(i) - x1(i) + 8;

    x1(i+1) = x1(i) + dt * dx1;
    x2(i+1) = x2(i) + dt * dx2;
end

```

B. The minimum stopping distance to avoid collision is 7.89 meters. Thus, my initial guess from Question A is not correct. The vehicles do not collide since the following car does not reach 8 meters.

CONCLUSION

Using the numerical methods implemented in Problems 1-3, we were able to visualize the outputs of initial value problems in real-world scenarios. Euler's Method allowed us to utilize step-size approximations for epidemic patterns and traffic patterns. We executed Runge-Kutta 4th Order to analyze the spread of an epidemic to achieve higher accuracy. We were able to change variables, such as parameters and initial conditions, to learn how these differences affect the

model as a whole. This project allowed us to implement numerical methods into relatable, everyday situations that we experience, and thus made sometimes overwhelming mathematics and theory into useful and practical tools.

REFERENCES

[1] *Differential equations — History & Overview*. (n.d.). <https://www.setzeus.com/community-blog-posts/differential-equations-history-overview>

[2] Wolfram Research, Inc. (n.d.). *Kermack-McKendrick Model -- from Wolfram MathWorld*. <https://mathworld.wolfram.com/Kermack-McKendrickModel.html>

[3] *Microscopic traffic simulation*. (n.d.). https://www.civil.iitb.ac.in/tvm/nptel/535_TrSim/web/web.html